MECH 191 — Metallurgy

Week 12

Machining

November 9, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

The Cutting Zone & Machining Processes

Shear plane, chip types, heat generation, and surface integrity — then turning, milling, drilling, and grinding

02

Cutting Tool Materials

HSS to CBN — hardness, heat resistance, toughness, coatings, and how to choose the right tool for the job

03

Surface Finish, Integrity & Machining Defects

Ra, Rz, residual stress, grinding burn, machining cracks, chatter — and how to detect each one

Metallography Lab — Module 11

Module 11 • Week 12

Machining

What to do this week:

Examine near-surface grain condition at 500×. Identify any subsurface deformation from sectioning.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimen A — 1018 steel

· Specimen B — 1045 steel

· Specimen H — 6061-T6 aluminum

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	**Unit 2** Steel & Alloys	**Unit 3** Testing & Quality	**Unit 4** Manufacturing	**Unit 5** Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12 ← here** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 12 — At a Glance

Topics

- The cutting zone — shear plane, primary and secondary deformation zones, chip types, heat distribution, and surface integrity
- Turning and milling — single-point vs. multi-tooth cutting, operations produced, surface finish formula, up vs. down milling
- Drilling and grinding — twist drill geometry, hole tolerance, grinding wheel types, grinding burn, and dressing
- Cutting tool materials — HSS, cemented carbide, coatings (TiN, TiAlN), ceramics, CBN, PCD, and selection criteria
- Surface finish — Ra and Rz parameters, typical values by process, connection to fatigue and sealing performance
- Machining defects — built-up edge, chatter, grinding burn (re-temper and re-hardening), machining cracks, and recast layer in EDM

Resources

Reference Material

Kalpakjian Ch. 21, 22 & 23
Fundamentals of Cutting · Tool Materials · Machining Processes

Visualizers

Cutting & Chip Formation
Turning & Milling — Sandvik Coromant Calculators

Surface Finish

$Ra = f^2 / (8r)$ · feed & nose radius control roughness in turning

Next: Week 13 — Powder Metallurgy

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What happens at the cutting zone during machining and why does it matter for part quality?
2. What are the main machining processes and what does each one produce?
3. What are the main cutting tool materials and how do you choose between them?
4. How is surface finish measured and why does it matter for performance?
5. What machining-related defects should a technician know how to recognize and detect?

The Cutting Zone — Where Dimensions and Surface Integrity Are Made

Three Deformation Zones

1 Primary Shear Zone

Chip forms along the shear plane ahead of the tool tip — most of the cutting energy consumed here. Severe plastic deformation; temperatures to 400–700°C.

2 Secondary Deformation Zone

Tool-chip interface — friction between chip and tool rake face generates significant heat. Drives tool wear and built-up edge formation at low speeds.

3 Tertiary Zone

Tool flank rubbing the new machined surface — contributes to work hardening and residual stress. Source of white layer damage if tool is worn.

Heat distribution: chip carries ~75–80% of cutting heat away from the part — which is why good chip control and coolant matter for surface integrity.

Chip Types & Cutting Parameters

Continuous Chip

Ductile materials at high speed with sharp tools — smooth ribbon; best surface finish. Chip control is a safety concern; chip breakers used.

Discontinuous Chip

Brittle materials (cast iron) or low speed — segments break off easily. Easier to control but may produce rougher surface.

Built-Up Edge (BUE)

Workpiece material welds to tool rake face at low speed — BUE fractures and adheres to machined surface; torn, rough finish. Prevented by increasing speed or improving lubrication.

Cutting Parameters

Cutting speed (Vc): Velocity of workpiece surface vs. tool (m/min) — higher speed → more heat, better finish above threshold

Feed (f): Tool advance per revolution or per tooth — controls chip thickness and surface roughness directly

Depth of cut (ap): Radial/axial depth per pass — controls material removal rate and cutting force

Turning & Milling — Single Point and Multi-Tooth Cutting

Turning

How it works

Workpiece rotates; single-point tool traverses — each revolution removes a helical chip strip

Operations

External turning (reducing diameter), facing (flat end face), boring (enlarging bore), threading, grooving, parting

Surface finish

Theoretical $Ra = f^2 / (8r)$ — finer feed and larger nose radius give smoother surface; finish turning achieves Ra 0.8–3.2 µm

Typical Ra values

Rough turning: 6.3–25 µm · Finish turning: 0.8–3.2 µm · Precision turning: 0.2–0.8 µm

CNC turning centers

Combine turning, milling, drilling in one setup — reduces handling, improves dimensional consistency for complex rotational parts

Milling

How it works

Rotating multi-tooth cutter removes material — each tooth takes a small chip per revolution; workpiece feeds past the cutter

Face milling

Large diameter cutter with inserts facing workpiece — generates flat surfaces on large areas; highly productive

End milling

Smaller cutter for slots, pockets, contours, and complex 3D surfaces — the workhorse of CNC machining centers

Up vs. Down milling

Up (conventional): chip starts thin, increases — less sensitive to backlash. Down (climb): chip starts thick, decreases — better finish, compressive surface residual stress, preferred on CNC

Surface finish

Finish milling: Ra 0.8–3.2 µm · Down milling produces compressive residual stress — beneficial for fatigue life

Drilling & Grinding — Holes and Finest Finishes

Drilling

- Two helical flutes carry chips out; two cutting edges at 118° point angle (135° for harder materials and stainless)
- Drilled hole: Ra 1.6–6.3 µm; tolerance H12–H13 — rough; follow with reaming (sizing) or boring (single-point) for precision bores
- Peck drilling: drill withdrawn periodically to break chips — used for deep holes to prevent chip packing and overheating
- Common errors: drill wandering at entry (use center drill first); work hardening at hole entrance in stainless (use sharp tools, adequate feed, no rubbing); BUE in aluminum (sharp HSS or uncoated carbide with cutting oil)

Grinding Types

Surface: Flat reference surfaces on precision components

Cylindrical: External diameters — shafts, bearing journals

Internal: Precision bores — hydraulic cylinders, bearing housings

Centerless: High-volume bar and tube — no centers required

Grinding Burn — The Most Serious Grinding Defect

Occurs when heat at the grinding zone exceeds the material's capacity to dissipate it — surface temperature rises above the tempering temperature of hardened steel.

Re-tempering Burn

Surface heated above tempering temperature — locally softened. Appears as dark band on Nital etch. Tensile residual stress introduced. Reduces fatigue life.

Re-hardening Burn

Surface heated above austenitising temperature then quenched by the bulk — untempered martensite white layer forms. Very brittle, high tensile residual stress. Most damaging type.

Detection

Grinding cracks: MT or PT — always reject in critical components. Burn: Nital etch (acid reveals re-tempered/re-hardened zones) or Barkhausen noise analysis. Wheel dressing restores sharpness and prevents burn from glazed wheels.

Cutting Tool Materials — From HSS to CBN

High Speed Steel (HSS)

Most versatile tool material — excellent toughness handles interrupted cuts. Suitable for complex geometries (drills, taps, form tools). Softens above ~600°C — limited to moderate cutting speeds. Used where toughness matters more than speed.

Cemented Carbide

Most widely used for production machining — significantly harder and more heat-resistant than HSS; allows much higher cutting speeds. Used as indexable inserts that are replaced rather than resharpened. Grades tailored to specific workpiece materials (P for steel, M for stainless, K for cast iron).

Coatings (TiN, TiAlN, Al_2O_3)

Deposited on carbide inserts — extend tool life 3–10× by reducing wear and friction. TiN (gold) is general purpose; TiAlN is superior at high temperatures; Al_2O_3 for high-speed steel machining. The familiar gold color of coated drills and inserts. Do not confuse coating type with base material.

Ceramics (Al_2O_3, Si_3N_4)

Used for high-speed finishing of hardened steels and cast irons where carbide wear is excessive. Extremely hard and heat-resistant but brittle — sensitive to interrupted cuts and vibration. Not suitable for general machining; specialist finishing tool.

CBN & PCD

CBN (cubic boron nitride) for hardened steels and superalloys — machines materials carbide cannot. PCD (polycrystalline diamond) for non-ferrous metals and composites — extreme hardness and tool life but cannot machine ferrous materials (carbon diffusion at cutting temperature). Both are used for high-precision finishing.

Surface Finish & Surface Integrity — More Than Just Ra

Roughness Parameters & Typical Values by Process

Ra (arithmetic mean roughness) — average deviation of profile from mean line; most widely specified parameter; lower Ra = smoother surface.
Rz (mean roughness depth) — average peak-to-valley height over several sampling lengths; more sensitive to deep valleys; more relevant for fatigue and sealing applications.

Process	Ra Range
Sand casting	6.3 – 25 µm
Hot rolling	12.5 – 50 µm
Rough turning/milling	3.2 – 12.5 µm
Finish turning/milling	0.8 – 3.2 µm
Grinding	0.025 – 0.8 µm
Lapping / Honing	0.012 – 0.4 µm

Surface Integrity

Surface integrity is broader than surface finish — it encompasses roughness, residual stress state, work hardening depth, and microstructural alteration. Specified for fatigue-critical aerospace, power generation, and medical components.

Compressive Residual Stress

Sharp tools, adequate coolant, down milling — compressive surface stress; beneficial for fatigue life. Shot peening deliberately introduces deep compressive stress after machining.

Tensile Residual Stress

Worn tools, excessive heat, aggressive grinding — tensile surface stress; dramatically reduces fatigue life. The primary damage mechanism in grinding burn.

Why Surface Finish Matters

Ra on a drawing is an engineering requirement, not an aesthetic — stress concentrations at surface peaks reduce fatigue life; rough bores leak; rough bearing surfaces wear rapidly.

Machining Defects — Recognition and Detection

Defect	Cause	Appearance	Location	Detection	Action
Built-Up Edge Marks	Workpiece material welds onto tool rake face at low speed — BUE fractures and adheres to machined surface	Torn, rough patches on machined surface; inconsistent surface finish	All over machined surface	**VT**	Increase cutting speed; use appropriate lubricant; replace dull tool
Chatter Marks	Vibration between tool and workpiece — tool overhang, fixture rigidity, or cutting parameter issue	Regular wave-like pattern on machined surface; audible during machining	Machined surface, oriented perpendicular to cutting direction	**VT, profilometer**	Reduce tool overhang; improve fixturing; adjust cutting parameters
Grinding Burn / Cracks	Insufficient coolant or aggressive grinding parameters — surface temperature exceeds tempering point	Discoloration (burn); network of fine cracks perpendicular or cross-hatch pattern	Machined surface — often invisible without etching	**MT, PT; Nital etch (dark/white zones); Barkhausen noise**	Reduce grinding depth/speed; improve coolant; dress wheel
Chatter Cracks / Machining Cracks	Severe thermal or mechanical conditions — most common in hardened steels during grinding	Fine cracks perpendicular to grinding direction or in a cross-hatch pattern	Ground surface of hardened components	**MT or PT — always a reject condition in critical components**	Reduce aggressiveness; correct coolant delivery; use softer wheel grade
Recast Layer (EDM)	EDM melts and re-solidifies thin surface layer — contains tensile residual stress and untempered martensite	White layer on cross-section metallography; no surface VT indication	All EDM-machined surfaces	**Metallography (cross-section); Nital etch**	Finish grind or polish to remove white layer in fatigue-critical applications

Technician's Perspective

Cutting parameters are quality parameters — not just productivity parameters.

Red Flag: Ground hardened shaft passes VT — but Nital etch reveals dark bands

A hardened steel shaft is ground to final diameter and passes visual and dimensional inspection. Nital acid etch reveals dark bands at regular intervals along the ground surface — re-tempered (softened) zones from grinding burn. The dark bands are locally softened martensite with tensile residual stress. This shaft must be rejected — the fatigue life has been compromised regardless of the dimensional compliance. Root cause: glazed grinding wheel or insufficient coolant. Do not accept any ground fatigue-critical component without specifying etch inspection.

Caution: Chatter marks on a milled bore evaluated as within Ra tolerance

A milled bore shows a wave-like surface pattern characteristic of chatter. Profilometer measurement shows Ra = 1.6 µm — within the specified limit. Do not accept on Ra alone. Chatter marks introduce cyclic residual stresses whose period matches the wave pattern — under pressure cycling they can nucleate fatigue cracks at a much lower stress than a uniformly rough surface. Flag the chatter pattern, identify the root cause (tool overhang, fixture rigidity, cutting parameters), correct it, and re-inspect.

Good Practice: Specifying NDT method on the machining drawing note

A bearing journal on a hardened shaft includes a drawing note: 'MT after final grind; Nital etch acceptance per AMS 2759/9.' The technician performs MT — no indications. Then applies Nital etch per the procedure — no re-tempered or re-hardened zones. Both steps are required; MT alone cannot detect grinding burn. The complete inspection record is signed off as the final release document. This is the correct sequence for any ground fatigue-critical surface.

Core Question Answers

What happens at the cutting zone and why does it matter for part quality?

Three deformation zones: primary shear zone (chip forms, most energy), secondary zone (tool-chip friction, heat), tertiary zone (flank rubbing, surface work hardening). Heat goes 75–80% into the chip, rest into tool and workpiece. Cutting parameters and tool condition determine surface integrity — residual stress, microstructural damage, and work hardening affect fatigue life and corrosion resistance.

What are the main machining processes and what does each produce?

Turning: workpiece rotates, single-point tool traverses — cylindrical surfaces, bores, threads. Milling: rotating multi-tooth cutter — flat surfaces, slots, pockets, complex 3D geometry. Drilling: twist drill — holes (rough, Ra 1.6–6.3 µm; ream or bore for precision). Grinding: abrasive wheel — finest finish (Ra 0.025–0.8 µm), hardened materials, finishing operation after turning or milling.

What are the main cutting tool materials and how do you choose between them?

HSS: toughest, resharpened, <600°C — for drills, taps, interrupted cuts at moderate speed. Carbide: hardest practical, indexed inserts, high-speed production — the default for most CNC work. Coatings (TiN, TiAlN): extend carbide life 3–10×. Ceramics: high-speed finishing of hard/cast iron, brittle. CBN: hardened steels and superalloys. PCD: non-ferrous, composites — cannot machine steel.

How is surface finish measured and why does it matter?

Ra = arithmetic mean roughness (µm); Rz = peak-to-valley depth. Ra on a drawing is a legal requirement — stress concentrations at surface peaks reduce fatigue life; rough bores leak and wear faster. Theoretical Ra for turning = $f^2/8r$ (feed^2 / 8 × nose radius). Surface integrity also encompasses residual stress, work hardening depth, and microstructural alteration — all affect fatigue performance.

What machining defects should a technician recognize and know how to detect?

Built-up edge marks (torn surface — increase speed/lubrication); chatter marks (wave pattern — fix vibration); grinding burn/re-tempering (dark Nital etch bands — tensile residual stress, reduced fatigue life — MT + etch); grinding cracks (network cracks — MT or PT, always reject); recast layer in EDM (white layer, tensile stress — remove by grinding in fatigue-critical parts).

Key Takeaways — Week 12: Machining

1. What happens at the cutting zone — shear deformation, heat generation, and tool-workpiece interaction — determines not just the dimensions of the machined part but its surface integrity, residual stress state, and fatigue performance. Cutting parameters and tool condition are quality parameters, not just productivity parameters.

2. Surface finish is a specified engineering requirement, not an aesthetic preference — Ra values on drawings must be verified. The connection between surface roughness, stress concentration, and fatigue life means that a surface that looks acceptable to the eye may still fail the specification and compromise service life.

3. Grinding burn and machining cracks are among the most serious manufacturing defects because they are often invisible to visual inspection and introduce tensile residual stresses and microstructural damage that dramatically reduce fatigue life. Recognizing the conditions that cause them and specifying MT, PT, or Nital etch inspection is essential knowledge for any quality technician.

Resources & What's Next

Week 12 Resources

Kalpakjian Ch. 21, 22 & 23
Fundamentals of Cutting · Tool Materials · Machining Processes — primary reading

Cutting & Chip Formation Visualizer
Shear plane, chip types, heat zones, BUE — interactive

Sandvik Coromant Calculators
Turning and milling cutting speed, feed, and Ra calculators — interactive

Nital Etch Acceptance (AMS 2759)
Standard for inspecting ground hardened steel surfaces for burn — reference

Barkhausen Noise Analysis
Non-contact grinding burn detection for production line use — reference

Week 13 — Powder Metallurgy

November 16, 2026 · Unit 4: Manufacturing continues

- What powder metallurgy is and why it enables materials impossible by any other route
- Powder production — water and gas atomisation, reduction, electrolytic deposition
- Compaction and sintering — green compact, neck formation, densification, HIP
- PM limitations and defects — density gradients, lamination cracks, blistering
- Process selection — how to choose between casting, forming, machining, and PM